

Lesson 5 Quantifiers

In this lesson, we continue to build our foundation, focusing on statements with **quantifiers**. A **universal statement** often begins with a **universal quantifier**, “**for all**” (*every, each, any*), denoted $\forall$: “For all x in set A , [*statement of x*]”:

$$\forall x \in A, s(x)$$

The following proposition is a universal statement:

$$\underbrace{\text{For all positive integers } n}_{\forall n \in \mathbb{N}}, \underbrace{n^2 \geq n}_{s(n)}$$

If a statement begins with an **existential quantifier**, such as “**for some**”, “**there exists**”, “**there is**”, “**there is at least one**”, denoted $\exists$, it is an **existence statement**,

$$\exists x \in A, s(x)$$

The following proposition is an existence statement.

$$\underbrace{\text{For some natural number } x}_{\exists x \in \mathbb{N}}, \underbrace{2^x < x^2}_{s(x)}$$

Another way to write the same statement is:

$$\underbrace{\text{There exists a natural number } x, \text{ such that}}_{\exists x \in \mathbb{N}}, \underbrace{2^x < x^2}_{s(x)}$$

In both cases we mean “there is *at least one* natural number x ”. If an existence statement begins with “there exists” or “there is”, add “such that” for readability.

It is very important that we learn to recognize if a statement is making a claim of a universal or existence nature, *whether the statement contains a quantifier or not*. The following statement does not contain a universal quantifier: “The sum of two even integers is even”, but its *arbitrary* character is clear: it means “The sum of **any** two even integers is even”, hence it is a *universal conditional*: “If (any) two integers are even, then their sum is even”, or another way to write it is:

For all integers x and y , if x and y are even, then $x + y$ is even.

A universal statement may take the form of a conditional. Many theorems are *universal conditionals*: a theorem is useful because of its *universal* character, and a theorem often presents a conclusion following some *conditions*.

The following statement is a universal statement: “A natural number is an integer.” It means “**Every** natural number is an integer”, different from the existence statement, “**There is a** natural number which is an integer.”

We may have more than one quantifier in a statement. The following statements are different, one is true while the other is false.

- a) For any integer x , there is an integer y such that $x + y = 0$.

$\forall x \in \mathbb{Z} (\exists y \in \mathbb{Z}, x + y = 0)$: **universal** containing an **existence** statement

- b) There is an integer x , such that for any integer y , $x + y = 0$.

$\exists x \in \mathbb{Z} (\forall y \in \mathbb{Z}, x + y = 0)$: **existence** containing a **universal** statement

Existence and **uniqueness** are two important concepts in math.

- Existence: “there is at least one”
- Uniqueness: “there is at most one”
- Unique existence: “there is a unique” or “there is exactly one”, denoted by “ $\exists!$ ”

The **Division Algorithm (DA)** is a unique existence theorem: Let $a \in \mathbb{Z}$ and $b \in \mathbb{N}$, then there are unique integers q (**quotient**) and r (**remainder**) such that

$$a = qb + r, \quad 0 \leq r < b$$

We first saw this in grade school: when we divide 9 by 4, we have $9 = 2(4) + 1$, so the quotient is 2 and the remainder is 1. It is *wrong* to say that $9 = 1(4) + 5$ and the remainder is 5, as $5 \not< 4$. Since DA is intuitive, we will treat it as an axiom, although it can be proved using more elementary notions (Lesson 10).

We move on to negation. To negate a universal statement, it suffices to find a **counterexample**, that is, an instance of x such that $s(x)$ is negated, so the negation of a universal statement is an existence statement and vice versa.

$$\sim(\forall x \in A, s(x)) = \exists x \in A, \sim s(x), \quad \sim(\exists x \in A, s(x)) = \forall x \in A, \sim s(x)$$

- a) For all integers x , $x^2 \geq x$.

Negation: There is an integer x such that $x^2 < x$.

- b) The sum of two even integers is even.

Negation: There are two even integers whose sum is not even.

- c) There is a natural number x , such that $2^x < x^2$.

Negation: For every natural number x , $2^x \geq x^2$.

- d) For every integer x , there is an integer y such that $x + y = 0$

Symbols are useful for multiple quantifiers. First, we negate the universal:

$$\sim(\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, x + y = 0) = \exists x \in \mathbb{Z}, \sim(\exists y \in \mathbb{Z}, x + y = 0), \text{ then}$$

$$\sim(\exists y \in \mathbb{Z}, x + y = 0) = \forall y \in \mathbb{Z}, x + y \neq 0, \text{ so the final negation is}$$

$$\exists x \in \mathbb{Z}, \forall y \in \mathbb{Z}, x + y \neq 0$$

There is an integer x such that for all integers y , the sum of x and y is nonzero.

Warning: the *location* of quantifiers and connectives can have dramatic effects!

The following two statements are NOT equivalent:

- Integers are not all even.
- All integers are not even.

(a) is NOT a universal statement, it is the negation of a universal, “Integers are all even”. It can also be written “Not all integers are even” or “There is an integer which is not even”, so it is an existence statement! Its negation is “Integers are all even”. (b) is a universal statement, its negation is the existence statement, “There is an integer which is even”.